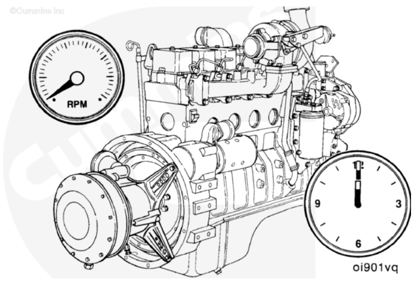
Trainings ▾

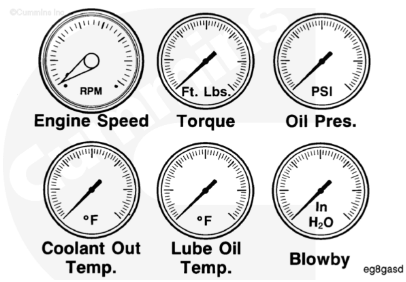
Test

The engine run-in period allows the tester to detect assembly errors and to make final adjustments needed for performance that meets specifications.

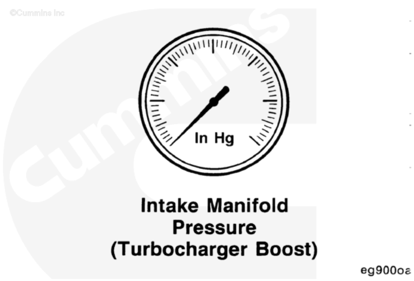
Note : The amount of time specified for the following engine run-in phases are minimums. Additional time can be used, if desired, at each phase except engine idle periods.



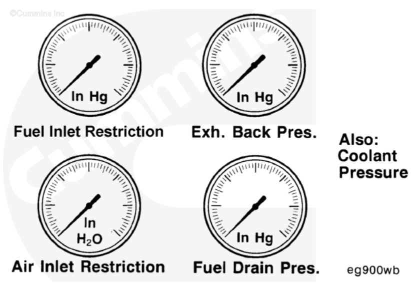
Measurements from these indicators and gauges **must** be observed closely during all phases of the engine run-in period. Refer to the appropriate sections for specifications and acceptable readings.



To evaluate the engine's performance correctly, this additional measurement **must** be observed during engine run-in phases.

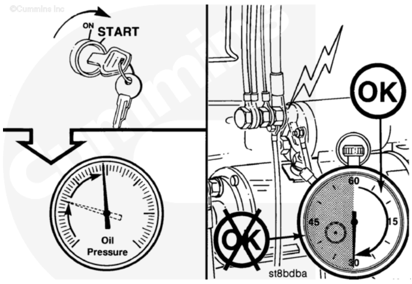


It is good practice to observe these measurements even if engine performance meets specifications. If engine performance does **not** meet specifications, these measurements can indicate possible reasons for nonperformance.



⚠ **CAUTION** ⚠

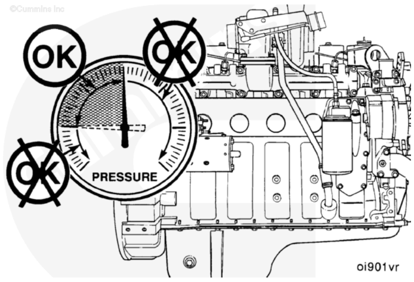
Do not crank the engine for more than 30 seconds. Excessive heat will damage the starting motor.



Crank the engine and observe the lubricating oil pressure when the engine starts. If the engine fails to start within 30 seconds, allow the starting motor to cool for two minutes before cranking the engine again.

⚠ CAUTION ⚠

If the lubricating oil pressure is not within specifications, shut off the engine immediately. Low lubricating oil pressure will cause severe engine damage.

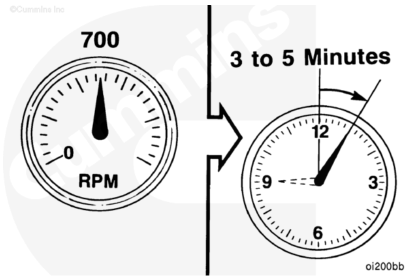


Engine lubricating oil pressure **must** be at least 69 kPa [10 psi] at 700 rpm.

Correct the problem if the lubricating oil pressure is **not** within specifications.

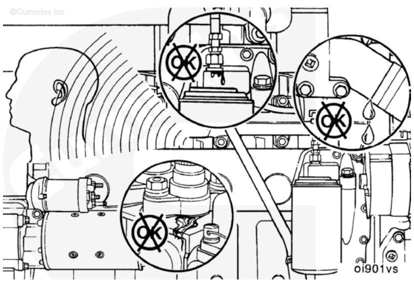
⚠ CAUTION ⚠

Do not operate the engine at idle speed longer than specified during engine run-in. Excessive carbon deposits will form in cylinders, causing damage to the engine.



Operate the engine at approximately 700 rpm for three to five minutes.

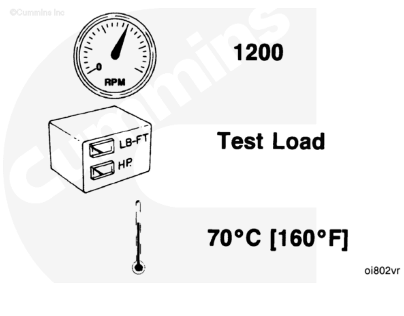
Listen for unusual noise; watch for coolant, fuel, and lubricating oil leaks.



Check for correct engine operation.

Note : Repair all leaks or component problems before continuing the engine run-in.

Move the throttle to obtain 1200-rpm engine speed, and set the test load to 25 percent of the rated load.



Operate the engine at this speed and load level until the coolant temperature is 70°C [158°F].

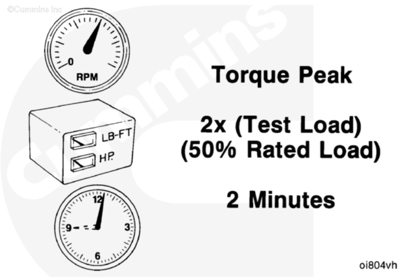
Check all gauges, and record the data.

Note : Do not proceed to the next step until a steady blowby reading is obtained.

Open the throttle to the speed at which peak torque occurs, and adjust the dynamometer load to 50 percent of torque peak load. Operate the engine at this speed and load level for two minutes.

Check all gauges, and record the data.

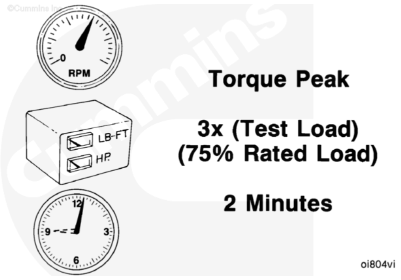
Note : Do not proceed to the next step until blowby is stable and within specification.



With the engine speed remaining at torque peak rpm, increase the dynamometer load to 75 percent of torque peak load. Operate the engine at this speed and load level for two minutes.

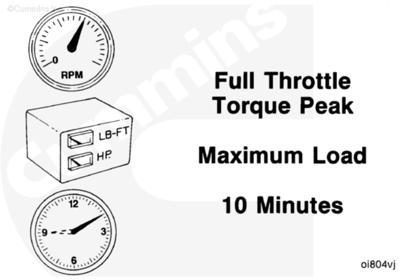
Check all gauges, and record the data.

Note : Do not proceed to the next step until blowby is stable and within specification.



Move the throttle lever to its full capacity in the opened position, and increase the dynamometer load until the engine speed is at torque peak rpm. Operate the engine at this speed and load level for ten minutes, or until the blowby becomes stable and within specification.

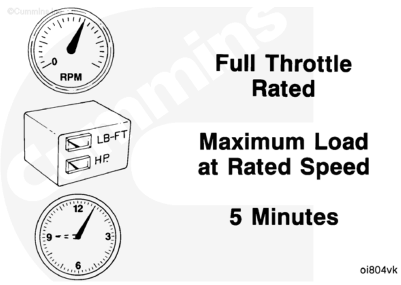
Check all gauges, and record the data.



Reduce the dynamometer load until the engine speed increases to the engine's rated rpm.

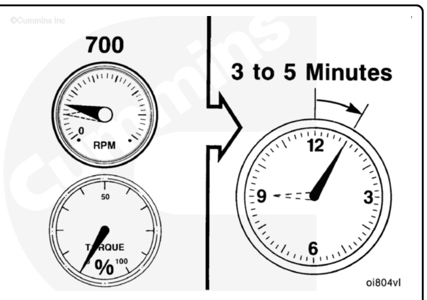
Operate the engine at rated rpm for five minutes.

Check all gauges, and record the data.



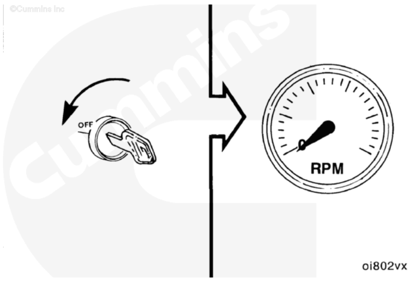
⚠ CAUTION ⚠

Shutting off the engine immediately after operating at full load will damage the turbocharger and internal components. Always allow the engine to cool before shutting it off.



Remove the dynamometer load completely, and operate the engine at 700 rpm for three to five minutes. This period will allow the turbocharger and other components to cool.

Shut off the engine.



Last Modified: 04-Feb-2004